Barangay VMS — Must-Haves and Requirement Alignment

Updated **28 October 2026** against the working tree on `dev-paul` (after `20449a7`). See the 28 October revision notes first, then the 25 October notes, for current behavior, audit disposition and verification limits. The supplied academic baseline remains unchanged.

Status key: **Implemented** means an identifiable code path exists; it does not claim deployment acceptance. **Verify in deployment** means credentials, storage, or an end-to-end acceptance check is still needed. **Open finding** means a confirmed defect prevents acceptance of the affected control. **Role alignment** means the capability exists but the responsible role differs from the supplied baseline and should be reviewed with the adviser.

Final-defense must-haves (28 October 2026)

The capabilities a panel should see working, with how to show each one and where it lives. Roles are the four in `backend-api/config/permissions.php`; each account holds **one** role. Demo accounts and sample data: Sample-Data.md.

#	Must-have	How to demonstrate it	Where it lives
1	**Registration and approval**	Register a new staff account (valid ID + photo + barangay staff code); show it inactive; approve it as the barangay Admin. For a barangay's first Admin, show the Super Admin pending queue and the emailed staff code that the Admin enters at sign-in.	Public `/register`; Users (Admin); Super Admin Pending Approvals; `AuthController`, `SuperAdminController`, `RegistrationSetting`
2	**Four roles, one role per account, least privilege**	Sign in as each role and show the different sidebars; try a forbidden action (e.g. Maintenance Personnel cannot report an issue; the API returns 403).	`config/permissions.php`; `modulesByRole` in `Workspace.jsx`; `user_roles` table
3	**Admin-created accounts with a temporary password**	Admin adds a user: a random temporary password is pre-filled (Regenerate available). Sign in as that user: the forced "Set your own password" modal appears and nothing else works until it is changed.	Users > Add User; `ChangePasswordRequired.jsx`; `users.must_change_password`; `EnsureUserIsActive`
4	**Vehicle registry**	Add a land and a water vehicle through the 3-step wizard (photo, documents, location); a new vehicle is Available / Good. Admin edits, archives, decommissions and restores.	Vehicle Management (Admin) / View Vehicles (Custodian, Maintenance Personnel); `FleetController`
5	**Vehicle types and locations**	Admin manages types and hubs; Custodian views types and adds a type or a hub (no delete); Maintenance Personnel see all barangay vehicles.	Vehicle Types, Vehicle Location; `HubController`, `FleetController`; `vehicle_type.*`, `hub.*`
6	**Condition monitoring that updates itself**	Propose a ticket on a Good vehicle and watch Condition Monitoring flip it to Needs Repair; after verified closure the vehicle returns to service.	Condition Monitoring; `conditionMonitoring.js`; `TicketController`
7	**Issue reporting (Custodian only)**	Custodian reports an issue with photos. Show that Admin and Maintenance Personnel have no Report action.	Report Vehicle Issue; `issue.create`; `vehicle_issue_reports`

#	Must-have	How to demonstrate it	Where it lives
8	**Ticket proposal with open-report linking**	Custodian starts a proposal on a vehicle that has open reports: only the "open reports" card shows; a report held by another ticket is greyed out; choose "None of these" to continue.	Report Vehicle Issue > propose; `ProposeTicketPage`; `GET /vehicles/{id}/open-issues`
9	**Approval workflow**	Admin reviews the proposal (default list: Proposals + Active), edits, assigns a mechanic per sub-issue, approves, declines or requests corrections. No direct ticket creation.	Maintenance Tickets; `ticket.approve/decline/request_corrections`
10	**Repair recording**	Mechanic opens My Work Orders and uses Record Repair Work: Standard Repair or Uses Parts From Another Vehicle, parts, evidence files; attachments then show on the ticket (View attachments).	My Work Orders; `LogRepairsPage`, `RepairFileCabinet`; `TicketController::logRepairs`
11	**Cannibalization from several donors**	In one repair add two donor vehicles (already-used vehicles are greyed out), each with parts; submit and show one issue report and a Needs Inspection flag per donor.	Cannibalization Management; `ticket_sub_issue_donor_parts.donor_vehicle_id`
12	**External-shop repair**	Custodian proposes an external repair, Admin only approves it; the Custodian logs the vehicle sent to the shop and its return, tests it and closes the ticket by verifying (or returns it for rework).	Ticket details; `external.handle`
13	**Independent verification and closure**	The assigned Custodian verifies (personal-test attestation) or returns the work for rework; approval closes the ticket; self-verification is refused. The ticket list shows the Title and Repair Type columns.	My Tasks (Custodian); `ticket.verify`
14	**Preventive maintenance**	Custodian books a schedule, Admin approves, the assigned mechanic completes it; recurring schedules renew; a due schedule becomes a ticket proposal.	Maintenance Schedule; `ConvertDueSchedulesToTickets`
15	**Maintenance records ledger**	Show completed repairs by source (In-House Repair, Sent to External Shop, Maintenance Schedule).	Maintenance Records
16	**History, reports and audit**	Vehicle History timeline; Admin generates reports and CSV; Activity Log shows who did what under which role.	Vehicle History, Reports, Activity Log
17	**Role dashboards**	Admin decisions queue, Custodian verification/readiness, Maintenance Personnel personal workload.	Dashboard
18	**Notifications and email**	Trigger a ticket event (in-app notification) and a registration event (email through Gmail SMTP with the VMS header logo and footer).	Notifications; `AccountMail`; `.env` `MAIL_*`

#	Must-have	How to demonstrate it	Where it lives
19	**Super Admin platform portal**	Barangay creation, registration codes, boundary review, pending approvals, concern reports with email replies, platform activity log, impersonation. Super Admin sees no fleet data.	Super Admin workspace; `RestrictSuperAdminScope`
20	**Data and file storage**	MySQL `barangay_vms_db` (47 tables); private Supabase Storage with signed links for photos, documents and evidence.	Backend `.env`; `ManagedFile`; ERD in the defense folder

Not in scope (do not claim them): dispatch, trips, Usage Log, direct Admin ticket creation, Admin verification of repairs, a self-service "forgot password" flow, and Maintenance Personnel issue reporting.

Current scope and audit status - 25 October 2026

This is a fleet **readiness and maintenance** system. Operational dispatch, a Dispatched availability state, trip tracking, Usage Log views and temporary-release workflows are outside the baseline and have been removed. Admin handling of an external-shop repair remains a maintenance workflow. Availability (Available / Under Maintenance / Inactive / Decommissioned), physical condition and verified readiness remain separate indicators.

The 24 October audit recorded six findings at `0027023`. Later integrated code rejects future completion dates. Other findings retain their specific acceptance retests rather than being automatically closed by integration. See current revision notes and section 8.

Working references for ERD and use-case revision

Use Requirements-Traceability.md for stable UC-01 through UC-18 identifiers, actor assignments, entity links and acceptance rules. Edit ERD.md and Use-Case-Diagram.md alongside this requirements source. The SVGs are current overview diagrams; PDFs and slides remain older exports. Requirements marked role alignment still need adviser agreement.

1. Platform requirements from the supplied document

Baseline requirement	Status	Current implementation / evidence
Web-based desktop platform	Implemented	React/Vite frontend and Laravel API; responsive layouts also support smaller screens.
Admin, Custodian, and Maintenance Personnel access	Implemented; open findings	Central abilities in backend-api/config/permissions.php, backend role/ownership guards, frontend module gates. Each account holds one role (`user_roles`, at most one row). Issue-detail visibility and global catalog writes have open authorization findings (AUD-24-02/03).
Organized, updated vehicle records	Implemented	FleetController vehicle endpoints, vehicle catalog and individual profiles, land/water type fields, photos and locations.
Maintenance records and schedules	Implemented	FleetController record/schedule endpoints; ticket repairs, recurrence, approval and scheduled-to-ticket processing.
Dashboard summaries of status and activities	Implemented; open finding	Distinct Admin decisions, Custodian inspection/verification, and personal Maintenance work dashboards. No invented trends or weather data. Readiness can remain fresh after parts removal; AUD-24-04 must be resolved.

Baseline requirement	Status	Current implementation / evidence
Secure cloud storage, backup and retrieval of images/files	Open finding; deployment checks (data is in MySQL `barangay_vms_db`)	Private managed storage and five-minute signed links exist, but an unrelated profile response can sign an inaccessible same-barangay file (AUD-24-01). Successful uploads, retrieval, retention and recovery depend on deployment credentials and storage configuration; backup/recovery has not been demonstrated by this UI change.

2. Administrator requirements

Supplied requirement	Status	Current behavior
Manage all vehicle records	Implemented	Register, edit and archive barangay-scoped vehicles. Vehicle registration is a standard Custodian duty under the current permission map; the obsolete delegation checkbox has been removed.
Manage vehicle categories/types	Open finding	Admin creates, edits and deletes; Custodian can view and add. Shared types and domains exist, but a barangay Admin can change types used by other barangays (AUD-24-03).
Update vehicle locations	Implemented	Vehicle locations and hub management with map views; occupied hubs cannot be deleted.
Monitor conditions needing inspection/repair	Implemented	Vehicle condition/readiness views and fleet-condition analytics; recording physical condition checks is Custodian-only.
Manage vehicle issue reports	Implemented	Review reports and linked tickets; dismiss obsolete reports with a reason. Custodians propose tickets.
Manage maintenance records	Implemented	Maintenance record oversight and repair-ticket history; ownership and verification guards still apply.
Schedule maintenance	Role alignment	Custodian books a schedule; Admin approves/declines and reassigns personnel. Admin does not originate a booking under the current permission map.
View vehicle history	Implemented	Automatic per-vehicle event timeline.
Generate vehicle, issue, maintenance and history reports	Implemented; open finding	Reports module plus relevant CSV/print exports. Validate report formats and neutralize formula-leading cells in all CSV exports (AUD-24-05).
Monitor logs and user activities	Implemented	Barangay-scoped activity log and platform-only Super Admin log.
Dashboard summaries and recent updates	Implemented	Four primary KPIs, decisions queue, ticket workload, fleet condition and concise coverage warnings. Details are linked from the queue.
Create user accounts	Implemented	Admin user.create permission; one role per account; a generated temporary password that the user must replace at first sign-in. Public signup additionally requires a registration code (except a barangay's first Admin), a valid ID and photo, and approval.

3. Custodian requirements

Supplied requirement	Status	Current behavior
View vehicle details and availability	Implemented; open finding	Vehicle catalog and profiles; availability, physical condition and verified readiness remain separate. A parts-removal event does not yet invalidate an earlier readiness pass (AUD-24-04).
Search/filter vehicles	Implemented	Search, filters, table/card views and column controls.
Report vehicle issues	Implemented (Custodian only)	Issue reporting with category, severity, description and linked proposal flow. The propose form asks about open reports first; reports already held by another ticket are greyed out.

Supplied requirement	Status	Current behavior
Upload issue photos	Verify in deployment	Issue-photo inputs/storage integration exist. Validate upload and authorized retrieval against configured cloud storage.
Monitor submitted reports/status updates	Open finding	Lists apply ownership restrictions, but direct issue-detail requests expose another staff member's hidden report and linked ticket details (AUD-24-02).
View ongoing/completed maintenance	Implemented	Maintenance ledger, ticket details, repair logs and verification history.
Dashboard of available vehicles, issues and maintenance	Implemented with role focus	Verification, readiness and unresolved issue KPIs; one tasks queue, one condition chart and readiness by vehicle type. Availability is accessible from Vehicles.

Additional current responsibilities: propose maintenance tickets, record condition/readiness checks, book schedules, view vehicle types and locations (add a type; add or edit a hub), log external-shop work (send to shop, receive back, rework), and independently verify completed repairs. Successful Custodian approval closes the ticket; Admin does not verify a repair.

4. Maintenance Personnel requirements

Supplied requirement	Status	Current behavior
View reported vehicle issues	Role alignment; open finding	Maintenance Personnel no longer file Issue Reports (Custodian only) and have no Issue Reports page; they see the problem through their assigned ticket and sub-issue. Direct issue-detail access still bypasses the list restriction (AUD-24-02).
Add maintenance records	Implemented with scope	Record repair work on assigned sub-issues and use permitted maintenance-record actions.
Update repair progress/completion	Implemented with scope	Assigned work logs and submission; a mechanic cannot independently verify or close their own repair.
Manage maintenance schedules	Role alignment	View and perform assigned scheduled work. Booking/editing is Custodian-owned; approval/reassignment is Admin-owned. Current completion validation rejects future Philippine dates; AUD-24-06 retains target-environment acceptance checks.
Add repair actions and remarks	Implemented	Repair log, parts, evidence, donor-part records and supporting notes. The current repair-log UI does not ask for Start/Completion dates.
Monitor upcoming maintenance	Implemented	Personal upcoming/overdue schedule KPIs and assigned-work queue.
View maintenance history	Implemented	Work tracker, completed repair history and permitted maintenance records.
View vehicles	Implemented	View Vehicles shows every vehicle in the barangay, not only assigned ones.
Dashboard of issues, records and vehicles needing attention	Implemented with personal scope	Three personal KPIs, assigned work, workload by actual ticket status and recent completed work. Fleet-wide analytics remain in authorized fleet modules.

5. Current workflow and safeguards

- ✓ Ticket proposals enter Pending Approval. Custodian proposes; Admin reviews/assigns; Maintenance Personnel repairs; assigned Custodian verifies and closes, or returns the work with a reason.
- ✓ Self-verification is blocked server-side, including multi-role accounts. Reassign an independent Custodian when needed; there is no Admin verification fallback.
- ✓ Cannibalized repairs record one or more donor vehicles and their parts; submitting files one follow-up issue report and a Needs Inspection flag per donor. The current flow does not require the retired Admin cannibalization approval step.

- ✓ External-shop work: the Admin only approves; the ticket's Custodian logs send and return, then tests and closes. Mechanics who did repair work cannot verify it.
- ✓ Schedule bookings require approval; recurrence and schedule-to-ticket processing remain intact.
- ✓ Vehicle-rooted tenant scopes and the Super Admin fleet allowlist exist.
- ☐ Close the issue-detail ownership, private-file signing and shared-catalog scope findings before treating access control as accepted (AUD-24-01/02/03).
- ☐ Invalidate stale readiness evidence after a parts-removal event (AUD-24-04).
- ✓ Last-Admin and open-work deactivation guards remain enforced.
- ✓ The dedicated switcher uses authorized account candidates and an automatically recorded access description; production tokens expire after 30 minutes and writes are restricted server-side. Local development has explicit expiry/write exceptions.
- ✓ The account menu identifies the current user and role. Ordinary support sessions offer End Impersonation; the dedicated switcher retains its account picker. Ending calls the existing logout endpoint for token revocation and audit logging before restoring the original session.
- ✓ Registration retains first-Administrator versus ordinary staff approval routing, registration-code checks, valid-ID and photo upload, and inactive-until-approved behavior.
- ✓ Admin-created accounts start with a temporary password and `must_change_password`; the API refuses every request except sign-out, "who am I" and the password change until it is changed.
- ✓ One role per account; creating or updating a user with more than one role is rejected.

6. UI requirements delivered in this revision

- ✓ Three focused fleet dashboards with real data, scoped assignments, task links and useful empty states.
- ✓ Restored Super Admin overview with three summary cards, administrative actions and a coverage chart.
- ✓ Flat Super Admin navigation and a dedicated Barangay Management screen; existing recovery, boundary review, registration codes and account controls remain accessible.
- ✓ Province → City/Municipality → Barangay → Staff impersonation cascade, using barangay IDs to distinguish duplicate names.
- ✓ Search/filter improvements for users, barangays, concerns and platform activity.
- ✓ Three-step registration form, wider desktop layout and readable introduction.
- ✓ Cleaner ticket-creation hierarchy and compact schedule summaries with full details under View.
- ✓ Shared eight-tooth blue/white SVG brand gear, transparent background, contrasting center ring, stationary wordmark and existing rotation timing. Reduced-motion preference is respected.

7. Acceptance checks and remaining work

Historical verification record (24 October): on revision `0027023`, the recorded full suite passed **480 backend tests (2,245 assertions)**; frontend lint, production build and targeted desktop/mobile merge checks passed. The 24 October investigation then reproduced the observed behavior with **six isolated API tests (36 assertions)** and a direct CSV-generation probe. Those investigation tests passed because they asserted the defective behavior; they are not evidence that the defects were fixed. No application fixes were made in this documentation update.

Acceptance remains open: resolve AUD-24-01 through AUD-24-06, retain regression tests that reject the bypasses, repeat affected workflows, and perform target-deployment storage/recovery acceptance. The SQLite removal-migration up/down/reapplication was verified on fresh and populated SQLite databases (the application database is now MySQL).

Historical verification notes - 15 to 21 October 2026

The results below describe their original revisions. Older statements that all findings were fixed do not close the findings recorded on 24 October.

Regression fix follow-up, 21 October 2026: all **22 audit findings have local fixes**. **422 backend tests pass (1,770 assertions)**, ESLint is clean, the production build passes without a chunk-size warning, seven geometry/calendar tests

pass, and **27 targeted browser assertions pass**. Storage now uses private managed objects and temporary signed links; the new migration was applied locally after a backup. Target-host bucket policy, persistent storage, legacy migration, and backup/restoration remain deployment acceptance checks. See the fix report for evidence and rollout instructions.

Regression audit, 21 October 2026: **415 backend tests passed (1,730 assertions)**, six geometry tests passed, and the frontend build passed. Browser/API regression testing recorded **22 open findings (4 High, 13 Medium, 5 Low)** across functional behavior, interface consistency, documentation, and tooling/performance. Frontend lint reports **59 errors and 18 warnings**. Registration, approval, internal/external repairs, rework, vehicle lifecycle, scheduled maintenance, reports, notifications, and support were exercised with isolated QA data. This is not deployment acceptance or exhaustive coverage of every control. See the current regression report for reproduction steps, evidence, previously fixed findings, and remaining coverage.

Registration geography, 15 October 2026: the cascading selectors now use **42,046 preloaded PSGC barangay references**, including Cebu City's **80 barangays**. Free-text barangay creation is rejected server-side; selected city/parent relationships are validated. Boundaries are resolved by PSGC code, with missing matches shown as unavailable instead of substituting another barangay. **403 backend tests passed (1,622 assertions)** after this change. See registration geography for source dates, boundary coverage, deployment steps, and evidence.

Boundary correction, 15 October 2026: Suba (Pob.), Bantayan now uses a detailed 1,281-vertex PSA/NAMRIA reference instead of the coarse triangle. Existing automatic references and new registrations share the correction; map rendering and location checks preserve holes and islands. **406 backend tests passed (1,649 assertions)** plus three geometry tests. See registration geography.

Suba verification follow-up: the government-hosted PSA/GeoRisk polygon agrees with the imported outline; the northern screenshot vicinity lies in Binaobao in that reference. Provider place labels are now optional, and the app labels the selected reference polygon internally. This verifies reference consistency, not current legal boundaries. See comparison and evidence. Five geometry tests and the build pass.

Boundary accuracy follow-up: imported polygons are now **unverified references**, including during registration. Super Admins can upload, preview, and confirm a replacement GeoJSON with its source recorded. Two automatic assignments were marked for review; live polygon coordinates and account/fleet/code data were unchanged. **Mantalongon's exact current geographic outline remains unresolved** pending a reliable replacement map. See boundary accuracy findings and fixes.

Regression audit, 15 October 2026: **397 backend tests passed (1,585 assertions)** and the frontend build passed. Browser/API checks and source review identified **19 findings**; REG-05, REG-11, REG-12, and REG-16 were subsequently fixed locally and browser-verified, leaving **15 open findings**. Baseline frontend lint reports **56 errors and 19 warnings**. See the regression report for reproduction steps, evidence, severity, and coverage limits. Passing backend tests do not close the remaining findings or establish deployment acceptance.

- ☐ Demonstrate cloud image/file upload, authorized retrieval, and backup/recovery in the target deployment.
- ☐ Obtain adviser review of the role-alignment differences in sections 2 and 4. These notes record current behavior; they do not claim approval to amend the original baseline.
- ☐ Complete user acceptance with representative accounts and real operational records, including an empty barangay and accounts that still carry two legacy roles.
- ☐ Confirm required generated report layouts and exports against the academic submission requirements.
- ☐ Maintain automated browser coverage for critical registration, verification and support flows. Backend feature tests exist in backend-api/tests/Feature; a passing frontend build alone is insufficient evidence of every workflow.

The original supplied document is not overwritten. Historical audit documents remain point-in-time records; their older Admin-confirmation, cannibalization-approval, and direct-ticket-creation descriptions should not be used as the current workflow contract.

Super Admin follow-up, 21 October 2026: all nine Super Admin audit findings and related retest edge cases were fixed locally. **433 backend tests (1,825 assertions), 98 API/browser scenarios, and 7 frontend tests passed; lint reports zero errors/warnings and the production build passes.** New barangays remain managed before onboarding, operational role changes protect assigned work, complete platform activity is searchable, and failures/retries are isolated by resource. Orphan recovery retains operational roles. Reference-only boundary onboarding remains

permitted when a live outline exists and no replacement is pending. See Super Admin fixes and verification for evidence, migration requirements, and coverage limits.

Later Super Admin recheck, 21 October 2026: the original 98 checks passed again, but expanded probes found **one new Low-severity issue: searching a concern by its displayed `CR-` reference returns no match**. Total API/browser result: 103 of 104 checks passed. Backend tests, frontend tests, lint and build still pass. See the latest recheck.

Concern-reference fix, 21 October 2026: SA-R01 is fixed. **All 104 Super Admin API/browser scenarios, 433 backend tests (1,825 assertions), and 7 frontend tests passed; lint and build passed.** Reference search also passed lowercase, whitespace, and numeric-ID checks. See the fix and complete recheck.

Second Super Admin recheck, 21 October 2026: **111 API/browser scenarios, 433 backend tests (1,825 assertions), and 7 frontend tests passed.** Lint and build passed; no new findings or browser runtime exceptions were recorded. The additional control checks cover combined filters, pagination state, sidebar state, code cancellation/copy, and invalid boundary files. See the latest recheck evidence.

8. Required acceptance criteria from the 24 October audit

ID	Priority / status	Requirement for acceptance
AUD-24-01	High - open	Validate parent-record access whenever a managed-file URL is issued or renewed. Echoing a file reference in profile/free-text data must not produce a usable signed URL for an inaccessible file.
AUD-24-02	High - open	Apply the same issue visibility policy to lists, direct details, nested tickets and attachments. A report hidden from an account must not be exposed by changing an ID in the URL.
AUD-24-03	High - open	Separate platform-wide catalog authority from barangay administration, or isolate tenant-owned catalogs. Renaming a shared value must not silently rewrite another barangay's completed records or inherited vehicle criticality.
AUD-24-04	High - open	Parts removal must invalidate readiness evidence recorded before that physical change. Require a fresh check without conflating availability, condition and readiness.
AUD-24-05	Medium - open	Export untrusted CSV cells as spreadsheet-safe text, including formula-leading characters and control characters. Test vehicle, user, report, schedule, history and work-tracker exports.
AUD-24-06	Implemented; acceptance retest required	Reject schedule completion dates later than the current accepted date. Invalid completion requests must not complete a schedule, create a ledger record or move its recurrence.
AUD-24-07	Low - documentation aligned	Current registration/import eligibility is Admin or Custodian; the legacy per-account registration flag is not an access gate. Baseline role changes still require adviser acceptance.
AUD-24-08	Low - documentation aligned	Current ticket cards show completion counts and stages; the ticket list also includes a progress bar and percentage. The old card progress ring is removed. Recording current behavior does not approve a new baseline requirement.

The requirements PDF is an earlier export of this Markdown reference and was not regenerated in this update. The original supplied baseline and earlier generated DOCX remain historical artifacts; neither is evidence that the new audit findings are resolved.